In [ ]:
```python
#1.Find the error in the following Python code:
import math
print(Math.sqrt(16))
```

In [1]:
```python
#Ans: The imported math module name should be lowercase.
import math
print(math.sqrt(16))
```

```
4.0
```

In [ ]:
```python
#2.Find the error in the following Python code:
def my_function():
    x += 5
    print(x)

x = 10
my_function()
```

In [3]:
```python
#Ans: x is modified inside the function without being declared as global.

def my_function():
    global x
    x += 5
    print(x)

x = 10
my_function()
```

```
15
```

In [ ]:
```python
#3.x = 5
if x = 10:
    print("x is 10")
```

In [4]:
```python
#Ans: for comparision operation we need to use "==".

x = 5
if x == 10:
    print("x is 10")
```

In [ ]:
```python
#4.
numbers = [5, 3, 8, 2]
print(numbers.sort())
```

In [5]:
```python
#Ans: .sort() modifies the list in place and returns None.

numbers = [5, 3, 8, 2]
numbers.sort()
print(numbers)
```

```
[2, 3, 5, 8]
```

In [ ]:
```
#5.
def square(n):
    n * n

print(square(4))
```

In [8]:
```
#Ans: In this function need to return a value or it prints none.

def square(n):
    return n * n

print(square(4))
```

```
16
```

In [ ]:
```
#6.
my_tuple = (1, 2, 3)
my_tuple[1] = 5
```

In [9]:
```
#Ans:Issue: Tuples are immutable and cannot be modified.

my_tuple = (1, 2, 3)
my_list = list(my_tuple)
my_list[1] = 5
modified_tuple = tuple(my_list)
print(modified_tuple)
```

```
(1, 5, 3)
```

In [ ]:
```
#7.
name = "Alice"
age = 25
print("My name is %s and I am %d years old" % (name))
```

In [10…
```
#Ans: The format placeholders are missing a required argument.
name = "Alice"
age = 25
print("My name is %s and I am %d years old" % (name, age))
```

```
My name is Alice and I am 25 years old
```

In [ ]:
```
#8.
a = [1, 2, 3]
b = [1, 2, 3]
if a is b:
    print("Lists are equal")
```

In [11…
```
#Ans:is checks for object identity, not value equality.
a = [1, 2, 3]
b = [1, 2, 3]
if a == b:
    print("Lists are equal")
```

```
Lists are equal
```

In [ ]:
```
#9.
for i in range(5, 1):
    print(i)
```

In [12…
```
#Ans: we need to apply -1 step for negative decrement value because the d

for i in range(5, 1, -1):
    print(i)
```

```
5
4
3
2
```

In [ ]:
```
#10.
student = {"name": "John", "age": 20}
print(student[name])
```

In [13…
```
#Ans: The key should be a string to get value.

student = {"name": "John", "age": 20}
print(student["name"])
```

```
John
```